EXERCISE 1

The function $f(x) = e^{2x} - 3$ cuts the x-axis at A and the y-axis at B.

1. Find the coordinates of A and B. [2 marks]

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2. Show algebraically that the function is increasing for all x. [3 marks]

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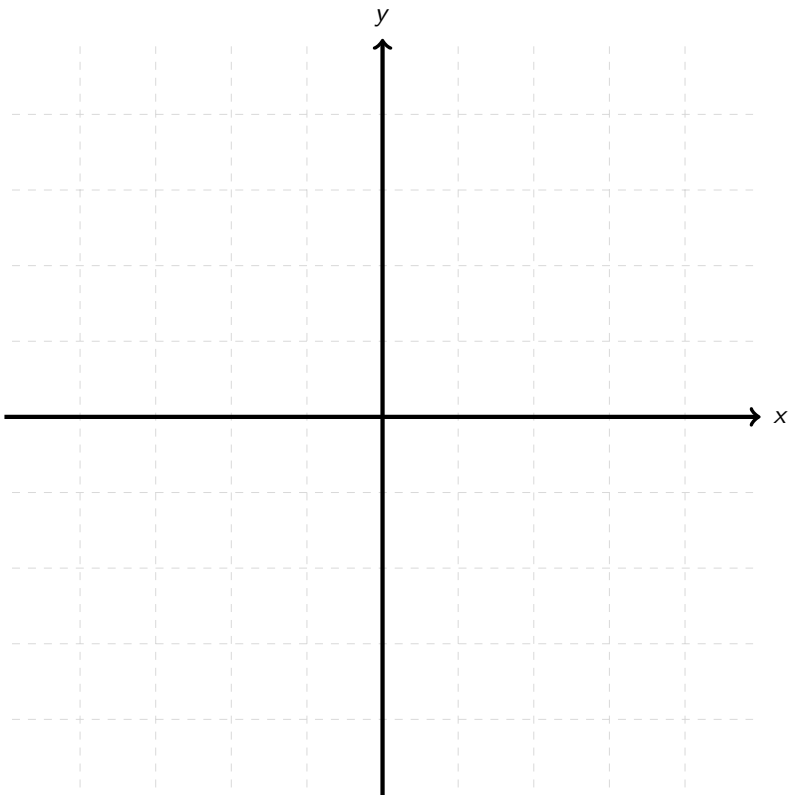
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3. Find $f''(x)$ and hence explain why $f(x)$ is concave up for all x. [2 marks]

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4. Use technology to help graph $y = e^{2x} - 3$. [3 marks]



EXERCISE 2 The derivative of a function f is given by

$$f'(x) = 4 + 2x - 3e^x, \qquad x \in \mathbb{R}.$$

1. Find the values of x for which f is decreasing. [3 marks]

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2. Find the values of x for which the graph of f is concave up. [3 marks]

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EXERCISE 3 The function f is defined by

$$f(x) = e^{2x}(3x - 4), \qquad x \in \mathbb{R}.$$

1. Find $f'(x)$. [3 marks]

2. Hence or otherwise, find the coordinates of the point on the graph of $y = f(x)$ where the tangent is parallel to the line $y = x$. [3 marks]

3. Find the equation of the tangent to the graph of f at this point. [3 marks]

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